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An Experimental Study of Turbocharged Hydrogen Fuelled Internal Combustion Engine

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ABSTRACT

Hydrogen is considered as one of the potential alternate fuel and when compared to other alternate fuels like CNG, LPG, Ethanol etc., it has unique properties due to absence of carbon. In the current work, Hydrogen engine of 2.5 L, four cylinder, spark ignited Turbocharged-Intercooled engine is developed for Mini Bus application. Multi-point fuel injection system is used for injecting the hydrogen in the intake manifold. Initially, boost simulation is performed to select the optimum compression ratio and turbocharger. The literature review has shown that in-order to get the minimum NOx emissions Hydrogen engines must be operated between equivalence ratios ranging from 0.5 to 0.6. In the present study, full throttle performance is conducted mainly with the above equivalence ratio range with minimum advance for Maximum Brake Torque (MBT) ignition timing. At each operating point, the performance, emissions and combustion parameters are recorded and analyzed in detail. It has been observed that a power of 66.7 kW @3200 rpm and a torque of 180 Nm @2000 rpm are achieved by using neat Hydrogen on Turbo charged engine. The base performance data is compared with CNG naturally aspirated engine (running using stoichiometric operation) and it has been observed that Hydrogen engine is having similar performance above 2000 rpm, however there is torque drop due to operation on Hydrogen which has limitation of the turbocharger to boost at lower engine speeds (running using equivalence ratio ranging from 0.5 to 0.6). Such lean operation also helps to avoid backfire which the hydrogen engine is quite vulnerable. It has also been observed that the combustion stability of the Hydrogen engine is very good at all Indicated Mean Effective Pressure (IMEP) values irrespective of the engine speed with a Coefficient of Variance (COV) around 1-2%.

Keywords: Hydrogen Engine, Combustion, Emissions, Performance, COV

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INTRODUCTION

The conventional petroleum base fuels such as Gasoline and Diesel are being used as a major source of fuels in transport vehicles. This sector is facing a potential threat due to uncertainty and interdependence of petroleum fuel availability and environmental issues like air pollution, global warming etc. These issues can be addressed to certain extent by use of alternate fuels. But most of the alternate fuels are hydrocarbon based that emit CO_2, which is a Greenhouse gas (GHG), may be at a lower rate than conventional fuels. Apart from CO_2 emissions, other pollutants such as CO, HC & NOx are emitted from the automobiles contribute to ambient air pollution. In order to reduce the CO_2 emission and ambient emissions, various fuels have been considered as substitutes to the conventional hydrocarbon based fuels.

The potential alternate fuels to replace them are mainly alcohols, Liquefied Petroleum Gas (LPG), Compressed Natural Gas (CNG), and Hydrogen (H_2), vegetable oils, bio gas, producer gas and Liquefied Natural Gas (LNG). Among all alternate fuels, Hydrogen (H_2) can be considered as a unique alternate fuel as it is a carbon less fuel and renewable fuel with capability to produce ultra-low emissions. Hydrogen gas, the simplest element in the universe, is one of the cleanest burning alternative fuel available and has the potential to eliminate all carbon emissions. Hydrogen can be adapted to both Spark Ignition (SI) engine and Compression Ignition (CI) engine. In SI engines, hydrogen can be used as a sole fuel, but in the case of CI engine dual fuelling technique can be adopted [2].

Since Hydrogen is a carbon free fuel, on burning hydrogen it will produces only water as its product. Hence the formation of HC, CO & CO_2 during the combustion can be completely avoided; however a

trace amount of these compounds may be formed due to the partial burning of lubricating oil in the combustion chamber. NOx is one of the major pollutants in hydrogen operated SI and CI engines [2]. NOx formed by the thermal dissociation and oxidation of nitrogen (N_2) in atmospheric air during combustion, is the only undesirable engine out emission from a neat hydrogen propelled engine [3]. But coupling this with unique characteristic of hydrogen, low lean-flammability limit, which allows stable combustion even at highly diluted condition [4], achieve near zero engine out emission.

Another characteristic of hydrogen is its tendency of pre-ignition and backfiring [5]. It is observed that these issues can be tackled if the proper selection of design parameters such as equivalence ratio, compression ratio is determined for stable engine operation. Hence in the current work, we chose to arrive at a best possible equivalence ratio and compression ratio to develop a practical & stable hydrogen engine which is targeted to be used on mini bus vehicle for transportation. The study emphasis on the process of development and on the testing of the hydrogen engine to understand the performance, emission and combustion characteristics.

METHODOLOGY

The following methodology is used in the present work.

We have checked in the literature the best possible region of equivalence ratio with which we can operate engine to get the minimum NOx emissions. Initial simulation was done with AVL boost simulation package to determine an optimum compression ratio and turbocharger configuration.

Hydrogen engine operating strategy is to determine the optimum equivalence ratio for minimal NOx emissions. Based on the optimum simulation data, a four cylinder Hydrogen fuelled engine is redesigned for Hydrogen from a base CNG engine.

Further, in this paper, emphasis is given on the hydrogen engine testing to understand the performance, emission and combustion characteristics.

HYDROGEN ENGINE OPERATING STRATEGY

The main parameters which need to be considered while operating a Hydrogen engine are performance and emissions. The Hydrogen engine should give best possible performance with lowest NOx emissions. Considering the learning from the previous work, if we reduce the combustion chamber flame temperature or in-cylinder thermal loading, NOx emissions from the Hydrogen engine reduces [6]. At the same time, it also reduces the heat losses from the combustion chamber walls, which will lead to increase in thermal efficiency. Due to this, risk of backfiring and pre-ignition also reduces as the reduction in combustion chamber temperature indirectly leads to reduction in hotspots inside the combustion chamber. The flame temperature can be reduced by using a very lean mixture.

To achieve lowest NOx emissions, we need to operate the engine in the equivalence ratio ranging from 0.5 to 0.6 which can be observed from Figure 1. At this equivalence ratio range, flame temperature gets reduced and it will lead to reduction in NOx emissions [7]. Hence we have decided to operate the engine in the above equivalence ratio range which is a conventional qualitative method of governing. This equivalence ratio range also reduces the risk of backfiring and pre-ignition.

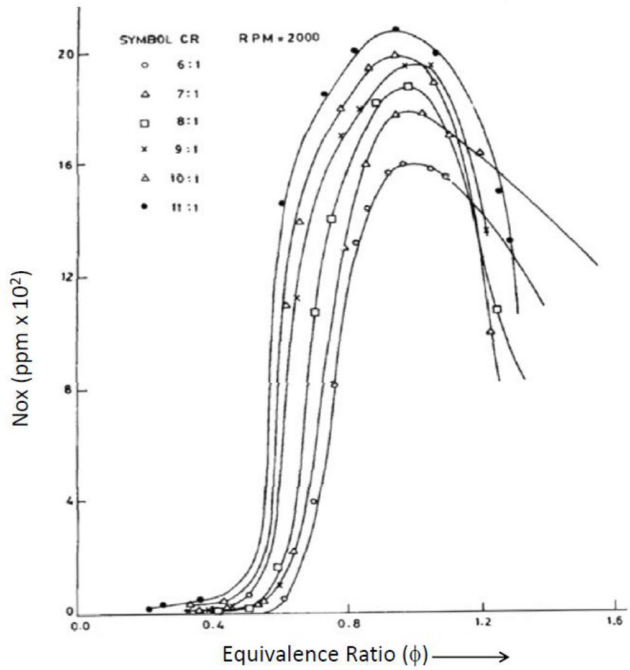

Figure 1. NOx Emission as a function of equivalence ratio for different Compression ratio [7]

AVL BOOST MODEL PREPARATION FOR HYDROGEN ENGINE

AVL Boost is one-dimensional gas exchange engine combustion simulation software, which can predict performance at steady state operating conditions. It can predict specific fuel consumption, power, torque, and mean effective pressures at full throttle and steady state operating conditions [8]. In the present study, AVL Boost is used to simulate the performance of Hydrogen engine.

The Hydrogen engine specifications are mentioned in the Table 1.

Table 1. Hydrogen Engine Specification

Base Specifications	
Number of Cylinder	4
Stroke	88.9 mm
Bore	100.3 mm
Number of Valves per cylinder	2
Fuel	Hydrogen
Aspiration	Turbocharged, Intercooled
Power	66.7 kW @ 3600 rpm
Torque	180 Nm @ 2000 rpm
Compression Ratio	12:1

The first step of simulation task is to create an accurate engine model which should consist of all the elements most similar to the existing physical engine. Figure 2 represents the engine model used for the current simulation.

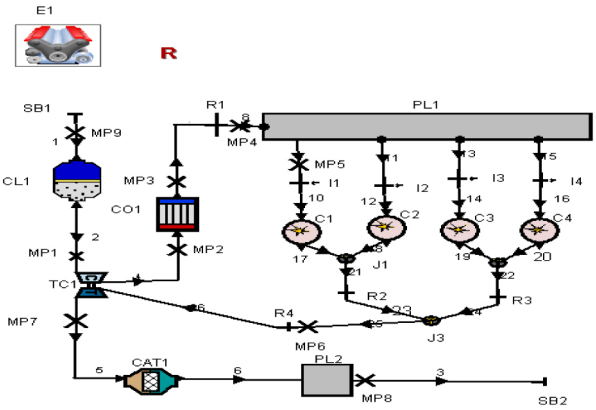

Figure 2. Boost Simulation Model

AVL Boost enables a detailed simulation of entire engine including the complete intake and exhaust system. Figure 2 shows the AVL Boost simulation model of the 2.5 L Hydrogen engine.

Description of the Model

The air is aspirated from system boundary SB1 through the intake snorkel pipe 1 to the air cleaner CL1. The cleaned air is sent to the turbocharger compressor via the pipe 2. TC1 is the turbocharger, from TC1 compressed air is sent to the intercooler CO1 via pipe 4. The intercooled air is passed through the restrictor R1 (which is modeled similar to the Throttle Body) to the intake manifold.

The intake manifold is split into two volumes. PL1 represents the plenum volume and the pipes 9, 11, 13 and 15 represent the runners for the cylinders C1 to C4 respectively. Pipes 10, 12, 14, 16 represent the intake ports for the cylinders C1 to C4 respectively. Injectors I1 to I4 are placed before the intake ports of the cylinders C1 to C4 respectively.

The exhaust ports are modeled in a similar way by the pipes 17, 18, 19 and 20. Before going into the exhaust manifold, the exhaust ports 17, 18 are connected to a common port 21 via junction J1 and the ports 19, 20 are branched to a common port 22 via junction J2 respectively. The exhaust ports 21 and 22 are connected at the flanges represented by restriction R2, R3 to the exhaust runner pipes 23 and 24 respectively. The exhaust runner pipes 23 and 24 are connected to junction J3 and from there they are sent to the exhaust outlet flange represented by restriction R4 through pipe 25. Exhaust gas enters into the turbocharger through pipe 26.

Exhaust gas expands in the turbocharger TC1 and goes to the catalytic converter CAT1 through pipe 5. From catalytic converter CAT1, the exhaust goes to the plenum PL2 (which is modeled similar to the silencer) through pipe 6. The exhaust goes finally let into atmosphere from plenum PL2 to ambient - system boundary SB2 through tail pipe 3.

Details of the Combustion Model

For thermodynamic calculations, rate of heat release is needed to be provided as input. The vibe function is often used to approximate the actual heat release characteristic of the engine [8].

$$\frac{dx}{d\propto} = \left(\frac{a}{\Delta\propto_c}\right).(m+1).y^m.e^{-a.y^{(m+1)}} \tag{1}$$

$$dx = \frac{dQ}{Q} \tag{2}$$

$$y = \frac{\propto - \propto_0}{\Delta\alpha_c} \tag{3}$$

where,

Q is the Total Fuel Input

α is the crank angle

α0 is the start of combustion

$\Delta\alpha_c$ is the Combustion duration

m is the shape parameter

a is the vibe parameter

The integral of the Vibe function gives the fraction of fuel mass which was burned since the start of combustion [7].

$$x = \int \frac{dx}{d\alpha}.d\alpha = 1 - e^{-a.y^{(m+1)}} \tag{4}$$

where,

x is the mass fraction burned

In the current work, Vibe 2 zone combustion model is used. The inputs for the Vibe 2 Zone combustion model are same as the Vibe function. However instead of one mass averaged temperature, two temperatures (burned and unburned zones) are calculated [9]. One more advantage of using this model is that this model also predicts the knocking characteristics of the engine, provided the actual heat release is described properly by the vibe function specified [9].

Vibe parameter (a) is specified as 6.9 as the combustion is assumed as complete combustion.

During our single cylinder hydrogen engine testing, pressure crank angle histories are captured. Using the same experimental pressure crank angle histories of our single cylinder hydrogen engine test data and the other parameters of the Vibe i.e., duration of combustion, shape parameter are calculated. Typical vibe parameters which are from that analysis are indicated below for reference [6].

During our previous work, a good correlation between the simulation work and experimental data is shown and hence, in the current work simulation is used to determine the compression ratio and turbocharger configuration. As the validation of Boost simulation data with respect to experimental data is already shown in our previous work [6], hence in this paper more emphasis is given on the experimental test data.

Table 2. Typical Vibe Parameters [6]

Speed (rpm)	Shape Parameter	Combustion Duration (^{0})
1600	2.3	20
2000	2.2	23.1
2400	2.24	28

Details of the Engine Friction Model

Engine friction adversely affects the maximum work output and fuel economy characteristics of an engine and directly accounts for much of the difference in fuel consumption between cold and full-warm engine operation [8]. For calculation of engine friction, the Shayler, Leong, Murphy - Model is used which is available in the Boost program. Based on the set of engine type and engine geometry related input data, Boost program calculates the frictional mean effective pressure (FMEP) of the engine and is provided as input to the simulation.

Details of the Knock Model

The knock model available in the Boost calculates the minimum octane number required for the engine operation free of knock [8].

The threshold for the onset of knock is exceeded if the integral [8].

$$\int_0^t \frac{1}{\tau_{id}(t)} dt \tag{5}$$

where,

T_{id} is the ignition delay at the unburned zone's condition is larger than one before the end of combustion is reached.

The ignition delay for the knock model depends on the octane number of the fuel and the gas condition. The equation for ignition delay is given below [8].

$$\tau_{id} = A.\left(\frac{ON}{100}\right)^a . p^{-n} . e^{\frac{B}{T}} \tag{6}$$

where,

T_{id} is the ignition delay in ms

ON octane number of the fuel

p is the pressure in atm

T is the temperature in K

A, a, n, B are the model constants

A = 17.68 ms

a = 3.402

n = 1.7

B = 3800 K

The approach for determining the model constants are by experimentally running the engine at different knocking conditions and obtaining the pressure crank angle histories at different conditions for knocking combustion. Fine tuning the above parameters in Boost case explorer in such a way that the predicted pressure curves are matching with the experimental pressure curves.

Since there are no model constants are available for hydrogen fuel and also no experimental cylinder pressure data with knocking conditions, the model constants which are specified in the Boost manual for the default condition are taken as the same for our simulation work and are as mentioned above.

Details of the Heat Transfer Model

The heat transfer model used for the simulation is Woschini model which is available in the Boost software and this model calculates the heat transfer rates to the piston, cylinder head, and liner [8]. Wall temperatures are provided as the input to the heat transfer model and the wall temperatures of piston, cylinder head, liner at TDC and liner at BDC are assumed as 585 K, 585 K, 585 K and 408 K respectively which are taken from our single cylinder hydrogen work as they provided a good correlation with the experimental work [6].

Details of the Species Model & Simulation Convergence Criteria

Classic species transport model is used to provide the thermodynamic properties.

The cycle calculation is done at a time step chosen automatically by the Boost software based on the CFL (Courant, Fredrichs and Lewy) stability criteria [8]. The cycle simulation is repeated till convergence to IMEP with specified error value is achieved.

Details of the Turbocharger Model

Boost provides two methods to model a Turbocharger.

1. Simplified Model
2. Full Model

Simplified Model is meant for steady state performance. This model provides further three modes for turbocharger simulation [8].

a. Boost Pressure Calculation
b. Turbine layout calculation
c. Waste-gate calculation

Full Model is used in unsteady engine operation [8].

In the current simulation work, simplified model is used along with the turbine layout calculation method to determine the turbocharger configuration.

Determination of Optimum Compression Ratio and Turbocharger Configuration using AVL Boost

Compression Ratio Determination

To determine the design feature of hydrogen engine, we need to determine the compression ratio. The simulation work is used to determine the compression ratio (CR) based on the Research Octane number (RON) values at different compression ratios and different speeds.

Simulation is done for different compression ratios ranging from 11:1 to 14:1 for the entire operating speed of the engine. It has been observed that for 11 compression ratio for the entire speed range of the engine, Research Octane Number (RON) value is in the range of 112 to 118. Similarly for 11.5 compression ratio - RON is in the range of 115 to 121, for 12 compression ratio - RON is in the range of 121 to 127, for 12.5 compression ratio - RON is in the range of 125 to 134, for 13 compression ratio - RON is in the range of 125 to 134, for 13.5 compression ratio - RON is in the range of 129 to 138, for 14 compression ratio - RON is in the range of 132 to 142.

To avoid knocking, the value of the compression ratio should be chosen in such a way that the obtained RON value should be below the RON value of 130. Hence, based on RON data for the entire range of engine speeds, a compression ratio of 12 meets the above criteria and is chosen for the engine with hydrogen.

Determination of Turbocharger Configuration

The turbocharger configuration is selected by using the following procedure. The required air mass flow rates and boost pressure are calculated at all engine speeds using the following equations [9].

$$\dot{m}_{air} = AFR.\frac{BMEP.V_d.N.BSFC}{2.2.16.10^9} \quad (7)$$

Where,

$\dot{m}_{air}$ is the required air flow rate in kg/s

AFR is the air fuel ratio

BMEP is the brake mean effective pressure in bar

BSFC is the brake specific fuel consumption in g/kW-h

N is the engine speed in rpm

V_d is the displacement volume in L

The required boost pressure is calculated by the below equation [9].

$$P_m = \frac{AFR.BMEP.BSFC.R.T_m}{\eta_v.3.6.10^9} \quad (8)$$

Where,

P_m is the inlet manifold pressure in bar

R is the gas constant of air in kJ/kg-K

T_m is the manifold temperature in deg K

η_v is the volumetric efficiency of the engine

After calculating the manifold boost pressure and air flow rates, the required pressure ratio of the compressor is calculated by considering the pressure drop across the intercooler and pressure drop across the air filter.

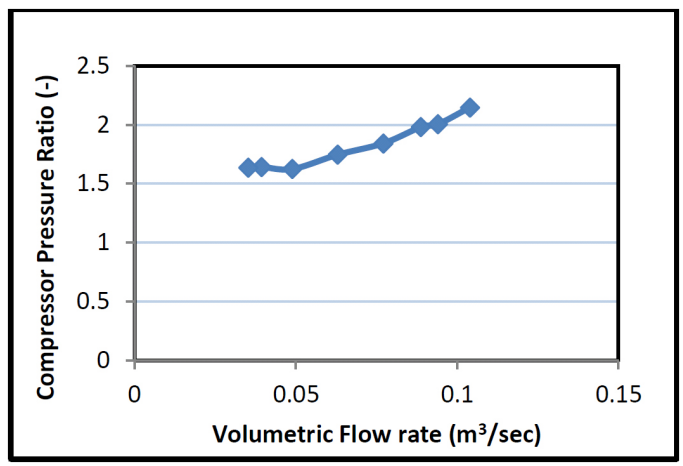

Figure 3. Compressor Pressure Ratio Vs Volumetric Flow Rate

Figure 3 shows the calculated Compressor pressure ratio with respect to volumetric flow rate of air. Using the calculated compressor pressure ratio, utilizing the turbine layout calculation in boost, the required size of the turbine is obtained. The calculated compressor pressure ratios are plotted on available compressor maps and a suitable compressor is selected. Similarly a suitable turbine is selected.

DEVELOPMENT OF PHYSICAL HYDROGEN ENGINE

The base engine used is a 4 cylinder CNG engine and further development is done on the engine to run on Hydrogen fuel. The details are as mentioned below.

Fuel Injection System

Timed manifold injection system is developed in place of the single point injection system. Various configurations of injection systems are studied and multi-point injection system is finalized. Normally in gasoline or CNG port injected engines, fuel can be injected before the inlet valve opens as the chance of pre-ignition or back fire is very less in those engines. But in a Hydrogen engine, injection of hydrogen before the opening of inlet valve can lead to back fire or pre-ignition. Hence, in our engine, hydrogen is injected after the inlet valve opening and this will reduce the chance of back fire. The induction of air initially will dilute the hot residual gases and cool hot spots if any.

Intake System

Same intake system of base CNG engine is used but instead of mechanical throttle body of CNG engine, an electronic throttle body is used to precisely control the air flow rate of the Hydrogen engine. Runner type intake manifold is designed and replaced the existing box type manifold of the CNG engine. The main reason for designing the runner type intake manifold is to have uniform distribution of air and to install the multi-point injection system of Hydrogen engine.

Compression Ratio

Based on the Boost simulation results, 12:1 compression ratio is finalized for the Hydrogen engine.

Turbocharger

Based on the Boost simulation results, suitable turbocharger configuration is selected.

Ignition System

Ignition system needs very low ignition energy to ignite Hydrogen fuel when compared to other fuels like CNG. As excessive energy can cause a hot spot on the spark plug electrode, which in turn will lead to backfire. Inductive type smart ignition coils are used in our application as they eliminate any residual voltage after the spark delivery. This is very much required for the hydrogen engine as any secondary unintended sparks caused due to residual voltage can lead to back fire.

Spark Plug

Cold rated spark plugs should be used for Hydrogen application. Platinum tips should not be used for Hydrogen application. A cold rated spark plug transfers the heat from the tip of the electrode to the cylinder head faster than the hot rated spark plugs which will reduce the chance of back fire for a cold rated spark plug compared to a hot rated spark plug which will reduce the chance of back fire for a cold rated spark plug compared to a hot rated spark plug. Since platinum acts as a catalyst causing the hydrogen to oxidize with air, hence platinum tipped spark plugs should not be used. Hence, in our current Hydrogen engine we have used cold rated Iridium tipped spark plugs. Copper electrode can cause corrosion due to the water vapor formation in Hydrogen engine and hence iridium tipped spark plugs is chosen for our engine.

Lambda Sensor

Narrowband switching type lambda sensor of base CNG engine is replaced with universal wideband oxygen sensor. As in our hydrogen engine we are operating in the equivalence ratio range of 0.5 ~ 0.6, narrowband oxygen sensor can't be used. As a narrowband oxygen sensor doesn't show the exact lambda value and it is widely used for lambda values of one operation.

Using the above developments a Hydrogen engine is built and tested for performance and emissions.

EXPERIMENTAL HYDROGEN ENGINE TESTING

Test Setup and Procedure

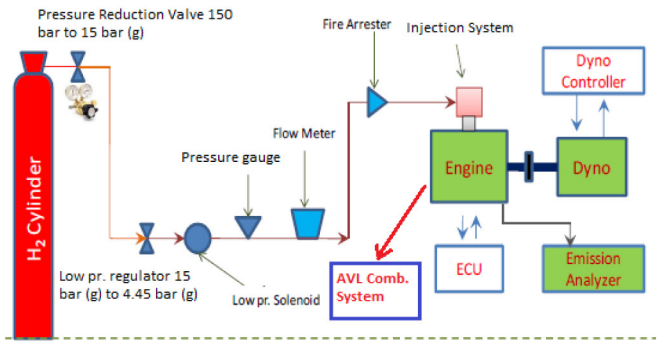

Figure 4. Experimental Setup

Figure 4 shows the experimental setup used for the Hydrogen engine testing. 12 Nos. of hydrogen cylinders are connected as a cascade and gas is supplied to a common manifold at a pressure of 150 bar (gauge). A pressure reduction valve reduces the 150 bar (gauge) pressure to 15 bar (gauge). A mechanical shut-off valve is installed between the common manifold and the pressure reduction valve to manually shut-off the gas supply to the engine.

Further, pressure is reduced to 4.45 bar (gauge) using a low pressure regulator. A low pressure solenoid is installed in the system which gets activated when we power on the system and allow the gas to flow further to the injection system. Low pressure solenoid is basically used for the safety of the system such that the hydrogen won't be flowing into the injection system during the power off condition of the engine.

A mechanical pressure gauge is installed in the system to check the actual pressure value during the engine testing. A flow meter is installed in the low pressure line to measure the mass flow rate of hydrogen. Flame arrester is used to arrest any fire which has happened outside the engine. A multi-point fuel injection system is used to supply the hydrogen fuel to the engine. ECU is programmed with different maps and control strategies to run the hydrogen engine at different operating conditions safely.

An eddy current dynamometer is used to load the hydrogen engine. A MEXA-720 direct installation gas analyzer is used for NOx measurement of the Hydrogen engine. An AVL Combustion pressure indication system is connected to the Hydrogen engine to process,

measure and analyze the cylinder pressure data. A direct mount type pressure transducer is installed into the combustion chamber and it is flush mounted in the combustion chamber. An optical encoder is installed on the crankshaft to measure the TDC point. Cylinder pressure data is measured along with TDC point using the cylinder pressure transducer and the optical encoder. The combustion pressure is analyzed and processed using the AVL Indicom combustion software. The combustion pressure data is analyzed to determine the Indicated Mean Effective Pressure (IMEP), peak pressure, combustion duration, Ignition delay and coefficient of variation of IMEP.

The coefficient of variation (COV) of IMEP for 100 consecutive cycles is calculated by using the below equation [10].

$$COV_{IMEP} = \frac{\sigma_{IMEP}}{\mu_{IMEP}}.100 \tag{9}$$

$$\sigma_{IMEP} = \sqrt{\frac{\sum_{n=1}^{100}\{IMEP(n) - \sigma_{IMEP}\}^2}{100}} \tag{10}$$

$$\mu_{IMEP} = \frac{\sum_{n=1}^{100} IMEP(n)}{100} \tag{11}$$

Where,

COV_{IMEP} is coefficient of variation of IMEP

n is the number of the cycle

σ_{IMEP} is the standard deviation of IMEP

μ_{IMEP} is the mean of IMEP

The heat release rate is calculated by the below equation [10].

$$\frac{dQ_c}{dt} = \frac{\gamma}{\gamma-1} P \frac{dV}{dt} + \frac{1}{\gamma-1} V \frac{dP}{dt} \tag{12}$$

where,

$\frac{dQ_C}{dT}$ is the heat release rate

γ is the ratio of specific heat

$\frac{dV}{dt}$ is the rate of change of Volume

$\frac{dP}{dt}$ is the rate of change of Pressure

In the present study, full throttle performance is conducted mainly with the above determined equivalence ratio range with minimum advance for Maximum Brake Torque (MBT). At each operating point the performance, emissions and combustion parameters are recorded and analyzed. The base performance data is compared with CNG engine (running using stoichiometric operation). At each operating point of the hydrogen engine, using the combustion pressure data the parameters like Indicated Mean Effective Pressure (IMEP), peak pressure, combustion duration, Ignition delay, heat release and coefficient of variation (COV) of IMEP are calculated.

Results and Discussion

The base vehicle CNG engine specification is as mentioned below.

Table 3. CNG Engine Specification

Base Specifications	
Number of Cylinder	4
Stroke	94 mm
Bore	94 mm
Number of Valves per cylinder	2
Fuel	CNG
Aspiration	Naturally Aspirated
Power	52 kW @ 3600 rpm
Torque	175 Nm @ 2000 rpm
Compression Ratio	12:1

Since the base vehicle used the above naturally aspirated CNG engine, we have used the same CNG engine to compare the power and torque of hydrogen engine. Hence, in the below discussion, we have initially shown the comparison for performance between CNG and Hydrogen engines and later more emphasis is shown on the Hydrogen engine testing data.

Performance and Emission

The Base CNG Engine is a naturally aspirated engine which is operated at Full throttle condition with Ignition timing maintained at MBT timing running using the stoichiometric air fuel ratio. The Hydrogen engine is turbocharged engine which is operated at Full Throttle condition with MBT timing using an equivalence ratio of 0.6 up to 1800 rpm, equivalence ratio of 0.5 from 2000 rpm to 3600 rpm.

It has been observed that turbocharged Hydrogen engine produced a power of 66.7kW at 3600 rpm and naturally aspirated CNG engine produced a power of 52kW at 3600 rpm which can be seen from Figure 5. Using Hydrogen fuel more power is achieved compared to CNG as the boost pressure ratio obtained is around 2.4 at 3600 rpm which can be observed from Figure 7 and also due to the boosted operation of the Hydrogen engine.

However, below 2000 rpm, it has been observed that the power achieved by the boosted Hydrogen engine is dropped compared to naturally aspirated CNG engine which can be seen from Figure 5. The power drop beyond 2000 rpm is due to low exhaust temperature produced by the Hydrogen engine which can be seen from Figure 7.

The exhaust temperature obtained before turbine by hydrogen engine at different speeds at full throttle condition is shown in Figure 7. The maximum exhaust temperature obtained from boosted Hydrogen operation is 873.16 K at 3600 rpm. But below 2000 rpm it has been

observed that the exhaust temperature is 723.16 K which is not sufficient to produce enough boost pressure ratio to increase the power at lower speeds below 2000 rpm.

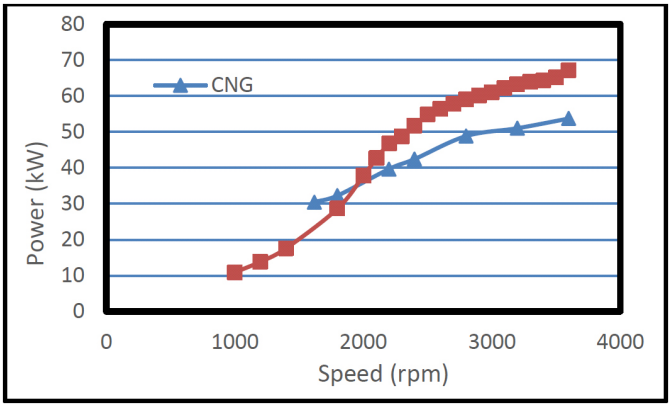

Figure 5. Speed Vs Torque of a Hydrogen Engine compared with a CNG Engine

Similarly the Hydrogen engine torque with respect to the CNG Engine Torque is shown in Figure 6. It has been observed that Hydrogen engine produced a Torque of 180 Nm at 2000 rpm whereas CNG Engine produced a torque of 175 Nm at 2000 rpm.

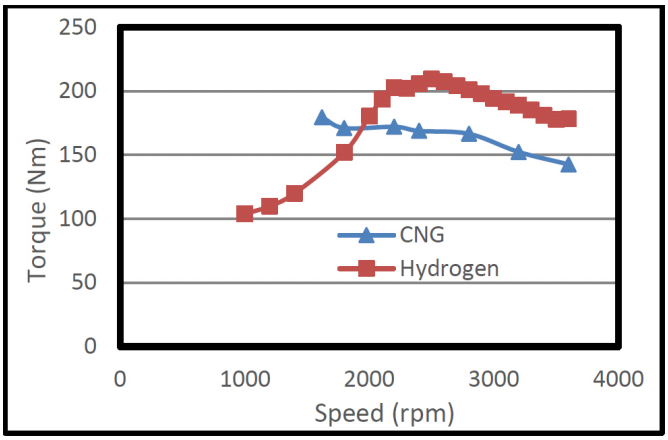

Figure 6. Speed Vs Power of a Hydrogen Engine compared with a CNG Engine

Below 2000 rpm, it has been observed that the torque produced by the boosted Hydrogen engine is less compared to naturally aspirated CNG engine whose trend is similar to the drop in power at lower speeds and this is due to low exhaust temperature of Hydrogen engine at lower speeds which can be seen from Figure 7. However in the vehicle, the actual torque delivered to the wheels at the lower speeds is compensated by changing the gears and drive ratios. It is also observed that the torque produced by Hydrogen engine above 2000 rpm is greater than the base engine and this is due to the boosted operation of the Hydrogen engine.

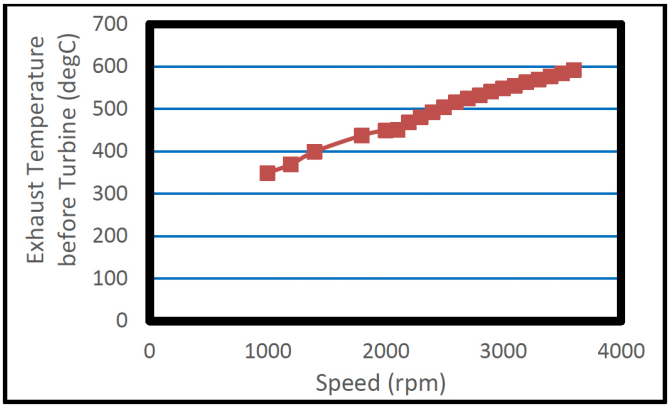

Figure 7. Exhaust temperature Vs Speed of a Hydrogen Engine

The variation of boost pressure of the Hydrogen engine with respect to mass flow rate of air which is shown in figure 8. It can be observed that the boost pressure ratio obtained at rated speed is 2.4 due to which more power is produced by Hydrogen engine.

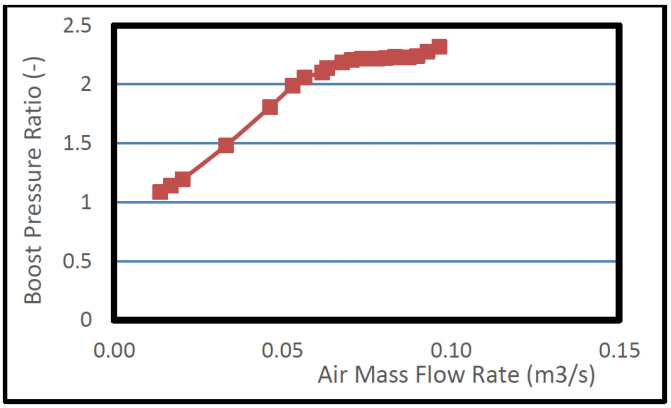

Figure 8. Pressure ratio Vs Volumetric flow rate of a Hydrogen Engine

As it can be seen from the Figure 8, the boost pressure ratio obtained at lower air flow rate of 0.025 m^3/s is 1.1 which is not sufficient to produce very high torque and this trend is matching with the drop in exhaust temperature shown in Figure 7.

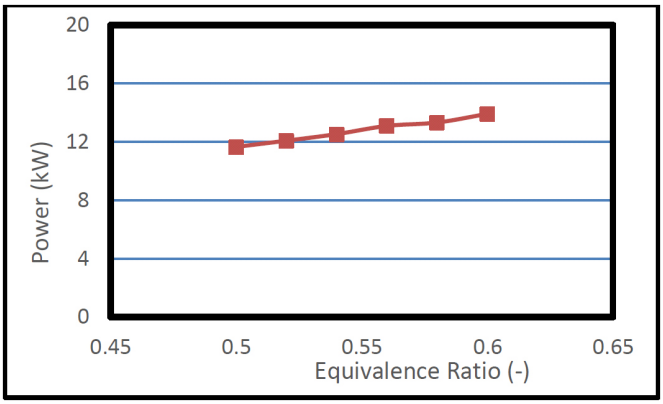

Figure 9. Power Vs Equivalence Ratio at 1200 rpm

To increase the power and torque at speed below 2000 rpm the engine is operated with an equivalence ratio of 0.6. As it can be seen from Figure 9 that the power can increase from 12kW to 14kW by increasing the equivalence ratio from 0.5 to 0.6 at 1200 rpm. Hence we have increased the equivalence at lower speeds to benefit from increase in equivalence ratio to 0.6 as further increase in equivalence ratio leads to increase in NOx emissions [7]. Also we have observed in our testing that after equivalence ratio of 0.65 engine tends to backfire.

Figure 10 shows the variation of Brake thermal efficiency with respect to speed of the turbocharged Hydrogen engine. It is observed that a maximum of 38% brake thermal efficiency is obtained at 2000 rpm. It is also observed the brake thermal efficiency obtained at all speed ranges is above 30% for the Hydrogen engine. The reason for this high thermal efficiency could be because of the faster combustion which leads to reduced heat losses to combustion chamber walls.

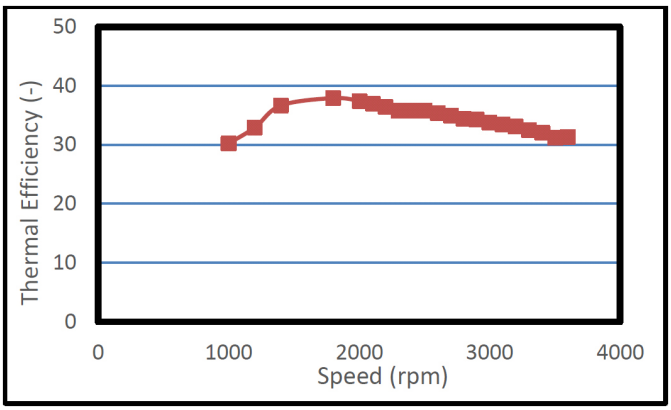

Figure 10. Brake Thermal Efficiency Vs Speed of a Hydrogen Engine

The variation of NOx obtained with respect to speed is shown in Figure 11. It has been observed that the NOx produced is around 500 ppm at rated speed and the NOx produced at lower speeds is in the range of 100 ppm.

It can be observed that the NOx produced by the Hydrogen is very low due to use of leaner equivalence ratio during the entire operating range of the engine. Lean hydrogen engine operation leads to reduced combustion chamber flame temperatures which causes the NOx formation rates to become slow.

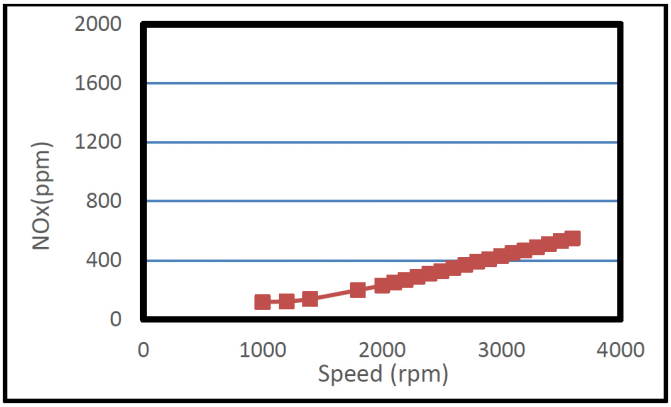

Figure 11. NOx emissions Vs Speed of a Hydrogen Engine

Combustion Analysis

The Cylinder Pressure (P) - Crank angle (θ) diagram for the Hydrogen engine at Maximum torque speed of 2000 rpm is shown in Figure 12. It has been observed that a peak pressure value of 67 bar is obtained at 2000 rpm. It can be observed that the Hydrogen combustion is faster from the rate of pressure rise curve shown in Figure18. The maximum rate of pressure rise at 2000 rpm is 2.6 bar/deg Crank angle (CA) which is shown in Figure 18.

The log P - log V variation for the Hydrogen engine at Maximum torque speed of 2000 rpm is shown in Figure 13. It can be observed from the log P - log V plot the start of combustion and the end of combustion using the compression and expansion processes. It can observed from the Figure 13 that the combustion duration for a Hydrogen engine is very small and hydrogen engine burns faster.

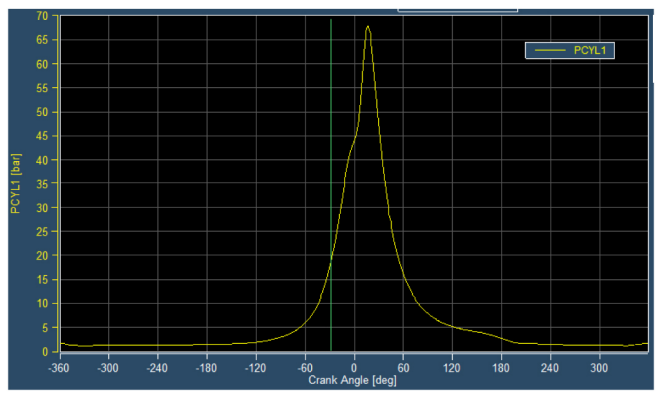

Figure 12. P-Theta Diagram at Max. Torque speed of 2000 rpm

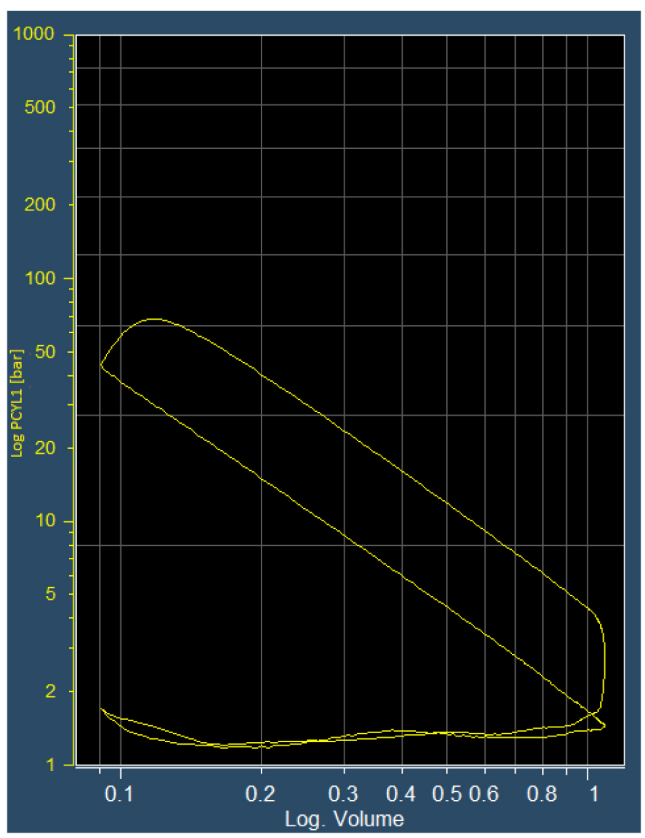

Figure 13. log P - log V at Max. Torque speed

The heat release rate variation with respect to crank angle for the Hydrogen engine at Maximum torque speed of 2000 rpm is shown in Figure 14. It can be observed from that curve that the rate of heat release in Hydrogen engine is very fast and all the heat is released in 180 CA duration at 2000 rpm. This faster heat release will obviously lead to better thermal efficiency as the heat losses to the combustion chamber will be lesser in Hydrogen engine due to the higher flame velocity of hydrogen.

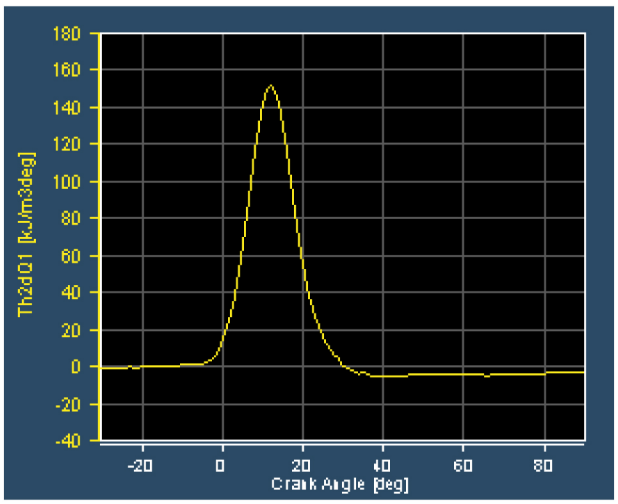

Figure 14. Heat Release rate Vs Crank Angle

The average peak pressure obtained from the Hydrogen engine at different speeds is shown in Figure 15. It can be observed from that a maximum of 98 bar peak firing pressure is obtained at 3600 rpm speed. The higher peak firing pressure of 98 bar at high loads lead to higher in-cylinder temperature which results in higher NOx values and this can be observed from Figure 11.

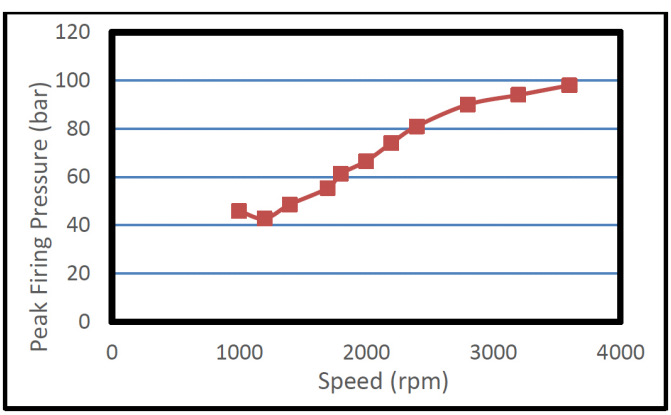

Figure 15. Average Peak Pressure Vs Speed

The variation of Ignition delay and Combustion duration with respect to speed are shown in Figure 16 and Figure 17. The Ignition delay is calculated as the duration between the start of Ignition timing and release of initial 5% of heat energy. The combustion duration is calculated as the duration between the initial 5% of heat energy release and 90% of heat energy release. It can be observed that the Ignition delay is around 110 crank angle at 3600 rpm and combustion duration is 300 crank angle at 3600 rpm.

Higher burning velocity of hydrogen compared to other hydrocarbon fuels will lead to faster combustion in hydrogen engine. The same trend has been observed in Figure 17 where the combustion duration is observed to be below 300 CA for all the speed ranges.

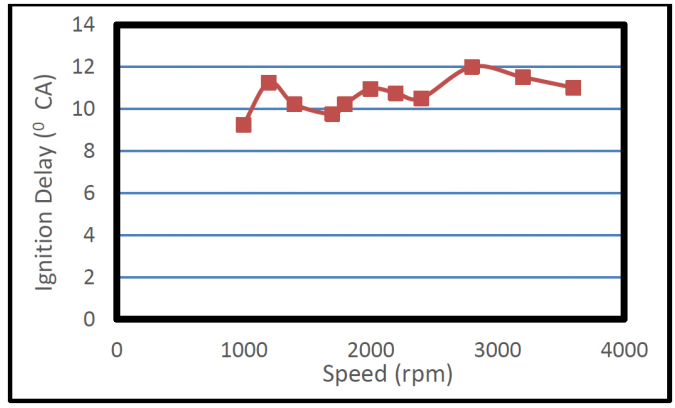

Figure 16. Ignition Delay Vs Speed

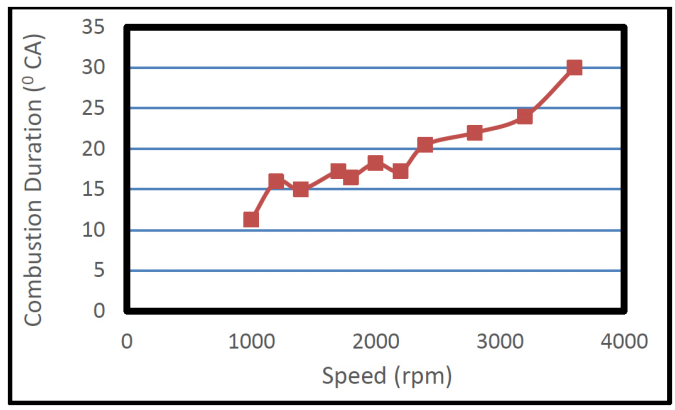

Figure 17. Combustion Duration Vs Speed

It has also been observed that the combustion duration of a lean burn Hydrogen engine is very much similar when compared with a stoichiometric CNG engine [11].

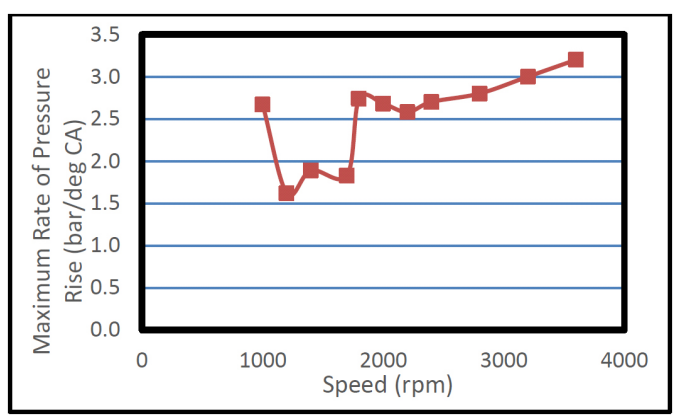

Figure 18. Maximum rate of Pressure rise Vs Speed

Figure 18 shows the variation of maximum rate of pressure rise with respect to speed. It can be observed that the maximum rate of pressure rise is 3.2 bar/^{0} CA at 3600 rpm. At mid-range of speeds the

maximum rate of pressure is low and as the engine speed increases the maximum rate of pressure rise increased at a very faster rate leading to faster combustion.

Figure 19 shows the variation of Coefficient of variation (COV) in Indicated Mean Effective Pressure (IMEP) with respect to speed. It can be observed that for the Hydrogen engine the combustion variation is very less and COV is below the value of 2% at all the speeds. It can be observed from the COV value that the Cycle-by-Cycle variations in combustion is very less for the Hydrogen engine even though we are operating the Hydrogen engine at very lean equivalence ratios.

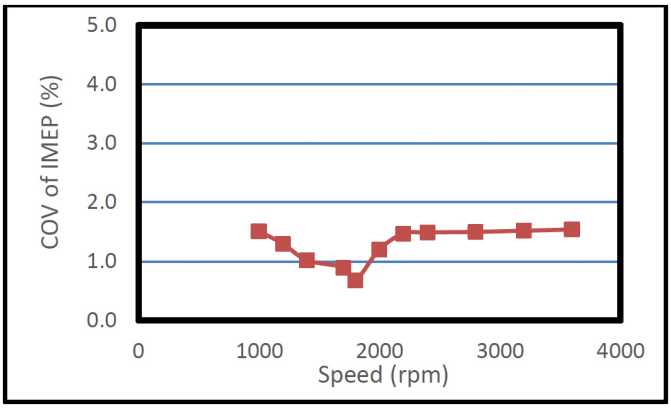

Figure 19. COV in IMEP Vs Speed

The experimental analysis shows that complexity of combustion in the hydrogen engine is mainly controlling the back fire and pre ignition tendency of the fuel. While it was demonstrated that the above factors can be controlled by selecting a combination of lean equivalence ratio coupled with controlled compression ratio which is substantially lower than the allowable limit as per hydrocarbon fuel Octane number. Another factor is the control of NOx emission. All the above factors have resulted in making the equivalence ratio to lean limit of 0.5 to 0.6. Such lean operation is affecting the power and torque rating of engine when compared to hydrocarbon fuel. Though the boosting by normal waste-gate turbocharger is regaining the power and torque, it is effective at the higher range of engine speeds whereas at lower range of speeds the lower exhaust temperature makes boosting ineffective and has shown lower torque values. This may have to be addressed by making specific designs of supercharger that can boost at lower speeds and independent of exhaust energy.

It is also seen that leaner operation is effectively managed in hydrogen engine because of the wide flammability limits of the fuel and due to the higher burning velocity of the hydrogen. This is reflected in the combustion analysis of the engine. The duration even at lean mixture is almost matching to stoichiometric combustion of hydrocarbon fuels. The study could demonstrate the positive advantages of hydrogen as a good fuel with no emission which makes it an attractive option in the future.

CONCLUSIONS

The goal of this work is to develop a hydrogen engine for Mini bus application and it has been achieved.

In-order to avoid back fire and to have optimum NOx level, an equivalence ratio of 0.5 to 0.6 is observed to be the right option for the Hydrogen engine. It has been observed that a compression ratio of 12:1 seems to give a safe operation of the Hydrogen engine. For the target application, the loss of torque due to lean operation in Hydrogen engine has to be compensated by turbocharging the engine, which can give a better performance compared to naturally aspirated CNG engine at higher loads.

It has been observed that the lower end torque of turbocharged Hydrogen engine is dropping drastically as the turbo cannot sustain energy requirement to provide required boosting due to low exhaust temperature of exhaust. It has been observed that the rate of combustion and Combustion duration of Hydrogen at lean equivalence ratio seems to be almost matching with hydrocarbon fuel at stoichiometric condition. It has also been observed that the cycle-by-cycle variation in combustion is well below the accepted limits for a hydrogen engine.

To summarize a Hydrogen fuel looks to be a good alternative fuel with near zero emissions compared to all hydrocarbon fuels.

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NOMENCLATURE

MBT - Maximum Brake Torque

IMEP - Indicate Mean Effective Pressure

COV - Coefficient of Variance

SB - System Boundary

CL - Air Cleaner

TC - Turbocharger

CO - Intercooler

C - Cylinder

R - Restrictor

PL - Plenum Volume

I - Injector

MP - Measuring point

J - Junction

CAT - Catalytic Converter

E - Engine

CA - Crank Angle

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